From 6% to 59%: A Data Analysis of Germany's Renewable Electricity Transition, 2000–2024, with a Comparison to France, Poland and Denmark

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Draft — not yet submitted. All results are computed from the published datasets cited below; analysis code is openly available (see Data Availability).

Abstract

Germany's electricity system has been rebuilt within a single generation. Using the Our World in Data energy dataset (compiled from Ember's Electricity Data Explorer and the Energy Institute Statistical Review) and the OWID CO_2 dataset, this study analyses annual data for 2000–2024 with descriptive statistics, growth rates and ordinary least-squares regression, complemented by a structured SQL analysis, and compares Germany with France, Poland and Denmark. The renewable share of German electricity rose from 6.2% in 2000 to 58.6% in 2024 (+2.05 percentage points per year, $R^2 = 0.95$), crossing 50% in 2023. Wind and solar supplied 80.8% of this

growth. Coal's share fell from 52.2% to 21.4% and nuclear generation ended completely (29.8% → 0%). The carbon intensity of electricity fell 41.3% and is strongly associated with the renewable share ($r = -0.95$; ≈ -4.6 gCO_2/kWh per percentage point). Two shocks — COVID-19 (demand −3.5% in 2020) and the 2022 energy crisis (a temporary coal rebound from 28.5% to 31.7%) — are clearly visible but did not reverse the trend. The comparison shows that decarbonisation strategy matters: Denmark leads on renewables (89.2%) and per-capita emission cuts (−55.5%), while France achieves the lowest carbon intensity (41 gCO_2/kWh) through nuclear power. A linear extrapolation of Germany's trend reaches the 80%-by-2030 target only in 2039, indicating that the target requires acceleration beyond the historical rate.

Keywords: renewable energy transition; Germany; electricity mix; carbon intensity; data analysis; Energiewende

1. Introduction

Few industrial countries have attempted to rebuild their electricity system as deliberately as Germany. The Renewable Energy Sources Act (EEG) of 2000 introduced guaranteed feed-in tariffs; the years after the 2011 Fukushima accident added a binding nuclear phase-out, completed in April 2023; and the amended EEG of 2023 set a target of 80% renewable electricity by 2030. This policy sequence makes Germany a natural case study for a simple empirical question: *how large, how fast, and driven by which sources was the change — and what did it do to emissions?*

This study answers that question with a reproducible data analysis of the

full period 2000–2024 and places Germany next to three instructive contrasts: Denmark, an early wind pioneer; France, which decarbonised mainly through nuclear power; and Poland, which remained coal-centred. Three contributions follow. First, the paper quantifies the German transition with a small set of transparent statistics rather than a narrative. Second, it separates the contribution of wind and solar from other renewables. Third, it tests how closely the fall in the carbon intensity of electricity tracks the renewable share — and documents where the association breaks, notably in the crisis year 2022.

2. Data and Methods

Data. All series come from two openly published datasets. The Our World in Data (OWID) energy dataset compiles electricity generation by source, source shares, electricity demand and the carbon intensity of electricity from Ember's Electricity Data Explorer and the Energy Institute Statistical Review of World Energy. CO_2 per capita comes from the OWID CO_2 dataset (Global Carbon Project). The analysis uses annual observations for Germany, France, Poland and Denmark, 2000–2024 (n = 25 per country). The extracted analysis table (100 rows × 23 variables) is published with the project.

Methods. The study combines three standard tools. (i) *Descriptive statistics and growth accounting:* changes in percentage points, multiples, and compound annual growth rates (CAGR) for generation by source. (ii) *Regression:* ordinary least squares (OLS) of the carbon intensity of

electricity on the renewable share for Germany (n = 25), reported with the Pearson correlation and R^2; and a linear time trend of the renewable share used for a deliberately naive projection of the 80% target year.

(iii) *SQL analysis:* the analysis table was loaded into a SQLite database and queried with nine documented queries covering shares, milestones (first year above 50% renewables; first year renewables exceeded coal), per-decade growth, the 2018–2024 shock window, and the cross-country per-capita comparison. All numbers in this paper were generated by the published code (`src/analysis.py`, `sql/queries.sql`); no figure is typed by hand. The regression estimates an association, not a causal effect; Section 5 discusses this limitation.

3. Results

3.1 The scale and pace of the transition

The renewable share of Germany's electricity rose from **6.22% in 2000 to 58.64% in 2024** — a 9.4-fold increase (Figure 1). The fitted linear trend is +2.05 percentage points per year (R^2 = 0.95), and the 50% threshold was crossed in **2023** (54.69%), nine years after Denmark (2014) but well ahead of France and Poland, which remain below 32%. Renewable generation overtook coal generation in **2019**. Extrapolating the 2000–2024 trend linearly, the share reaches 80% only in **2039** — nine years after the 2030 target date. The target therefore implies an acceleration relative to the historical trend, not its continuation.

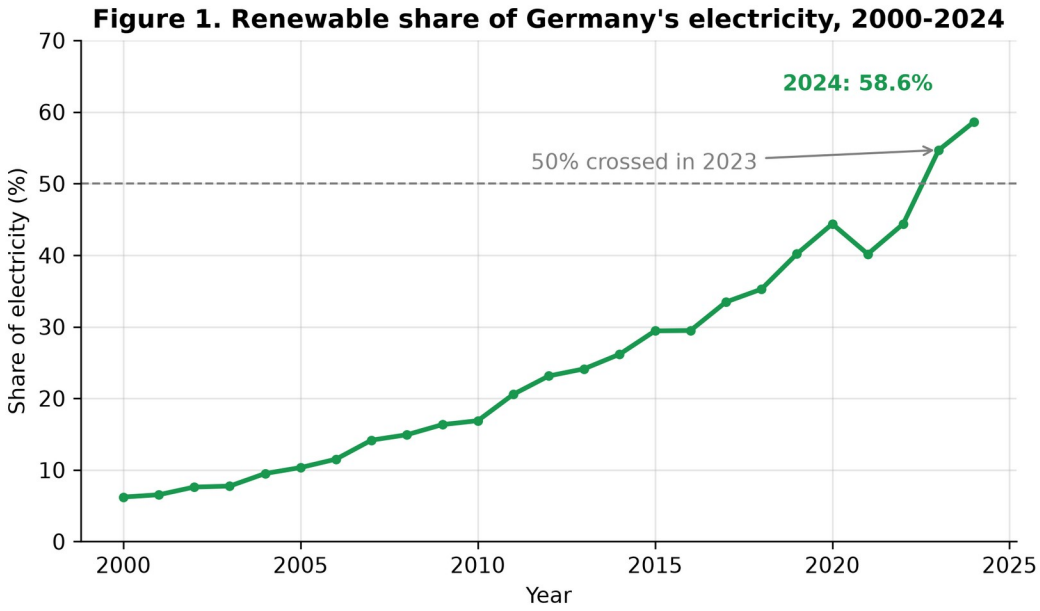


Figure 1. Renewable share of Germany's electricity, 2000–2024.

3.2 Wind and solar supplied four-fifths of the growth

Total renewable generation grew from 35.4 TWh (2000) to 290.9 TWh (2024).

Wind and solar together account for 80.8% of that increase (Figure 3).

Wind generation rose from 9.35 TWh to 141.6 TWh (CAGR 12.0%), reaching a 28.6% share of all electricity in 2024. Solar grew fastest: from 0.10 TWh in 2001 (the first non-zero year) to 74.13 TWh in 2024 (CAGR 33.3%), a 14.9% share. Hydropower, the dominant renewable of 2000, grew slowly, so the transition is overwhelmingly a wind-and-solar story.

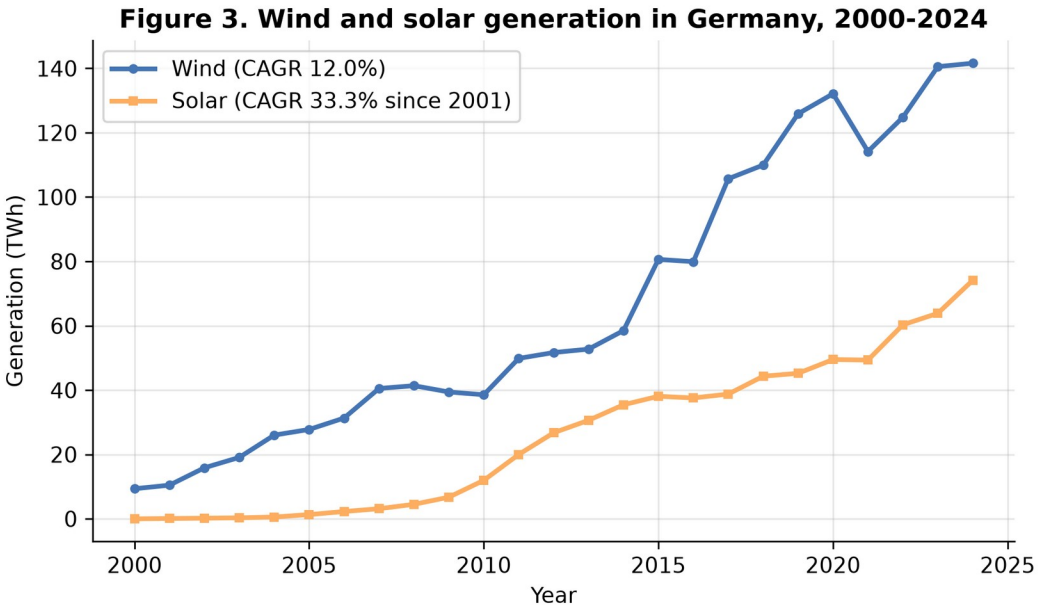


Figure 3. Wind and solar generation in Germany, 2000–2024.

3.3 Coal decline and the completed nuclear exit

The mirror image of the renewable rise is the fall of the incumbents (Figure 2). Coal's share dropped from **52.15% (2000) to 21.44% (2024)**, and coal generation fell in absolute terms from 296.7 TWh to 106.3 TWh. Nuclear power declined from 29.81% of generation in 2000 to 6.12% in 2022 and **0% in 2024**, following the completion of the phase-out in April 2023. Gas partly filled the gap, rising from 8.65% to 15.81%.

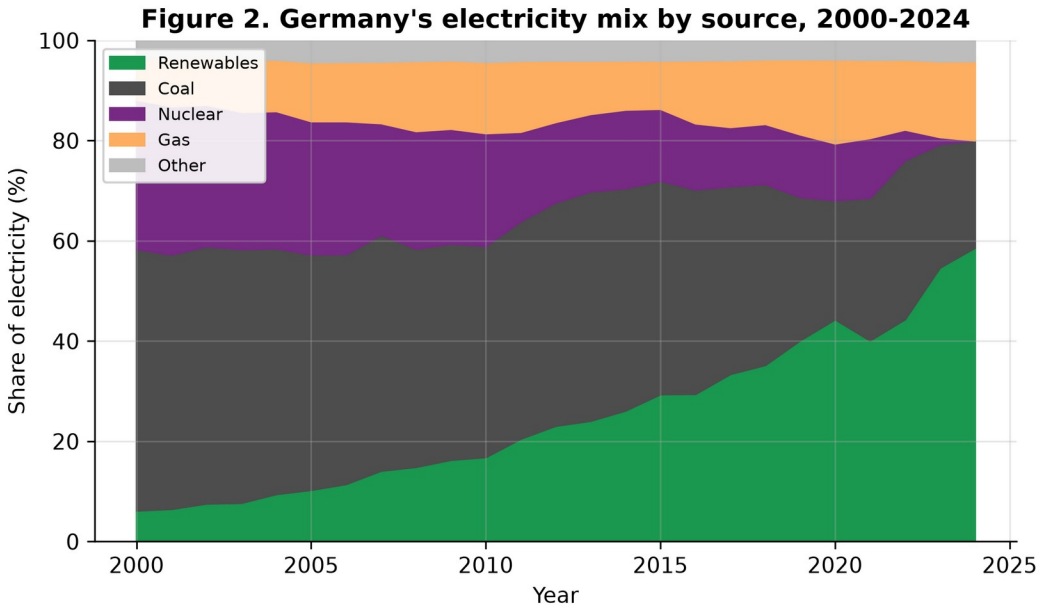


Figure 2. Germany's electricity mix by source, 2000–2024.

3.4 Emissions: a strong association, with one visible exception

The carbon intensity of German electricity fell from **573.2 gCO_2/kWh in 2000 to 336.4 gCO_2/kWh in 2024 (−41.3%)**. Across the 25 annual observations it is strongly negatively associated with the renewable share (Figure 4): Pearson r = −0.95, R^2 = 0.91, with an OLS slope of **−4.56 gCO_2/kWh per additional percentage point of renewable share**. CO_2 emissions per capita fell from 10.99 t to 6.79 t (−38.2%) over the same period (Figure 6).

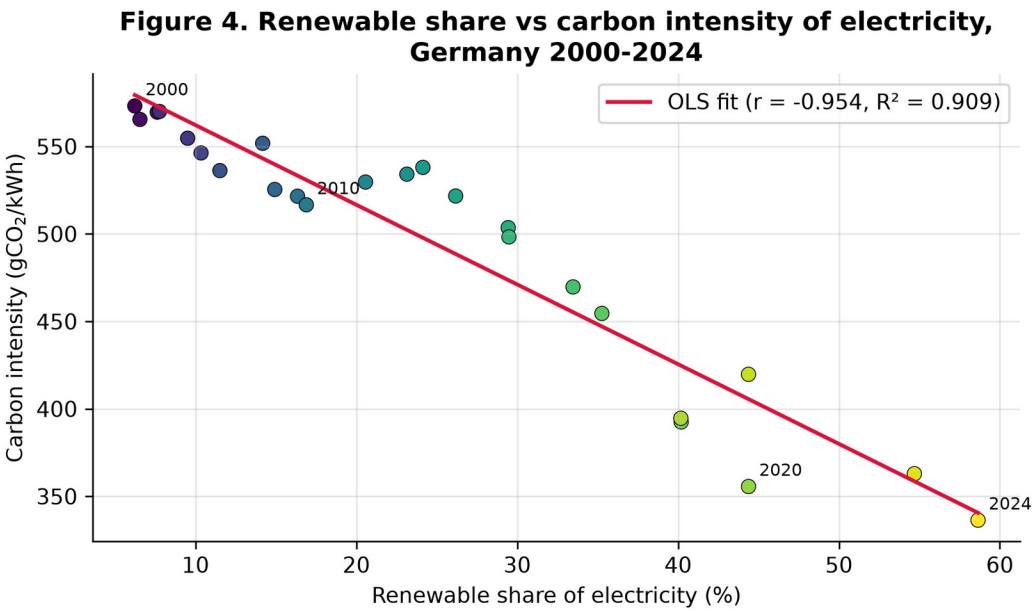


Figure 4. Renewable share vs carbon intensity of electricity, Germany 2000–2024. Colour scale: year.

3.5 Two shocks: COVID-19 and the 2022 energy crisis

The annual data register both shocks cleanly. Electricity demand fell **3.5% in 2020** (567.4 → 547.7 TWh) during COVID-19, then partly recovered in 2021 (+2.3%). In **2022**, after the cut in Russian pipeline gas, the gas share of generation fell from 15.60% to 13.95% while the coal share **rebounded from 28.45% to 31.74%** and demand fell 3.7% — a one-year, crisis-driven step backwards inside the long trend. By 2024 the coal share had fallen to a period low of 21.44% and the renewable share reached its period high, indicating the rebound was temporary.

3.6 Comparison: three different decarbonisation strategies

The 2024 comparison (Figure 5, Table 1) shows that a low-carbon system can be reached by different routes, at different speeds.

Countr	Renew	Renew	Coal	Nuclea	Carbon	CO_2 per	Change

y	ables 2000 (%)	ables 2024 (%)	2024 (%)	r 2024 (%)	intensity 2024 (gCO_2/kWh)	capita 2000 → 2024 (t)	(%)
Germany	6.22	58.64	21.44	0.00	336.4	10.99 → 6.79	−38.2
France	12.74	27.18	0.18	67.72	40.5	6.86 → 3.93	−42.8
Poland	1.63	31.15	54.32	0.00	608.2	8.29 → 7.23	−12.9
Denmark	15.54	89.19	4.99	0.00	131.8	10.17 → 4.52	−55.5

Table 1. Electricity transition indicators, 2000–2024. Sources: OWID energy and CO_2 datasets; values computed by the project code.

Denmark, starting earliest, reached 89.19% renewables and the largest per-capita emission cut (−55.5%). France has by far the cleanest electricity (40.5 gCO_2/kWh) despite a modest renewable share, because 67.72% of its generation is nuclear. Poland illustrates the cost of delay: still 54.32% coal, the highest carbon intensity (608.2 gCO_2/kWh) and the smallest per-capita reduction (−12.9%). Germany sits between: the fastest large-system renewable build-out of the four, but a carbon intensity still 8 times France's because one-fifth of its electricity still comes from coal.

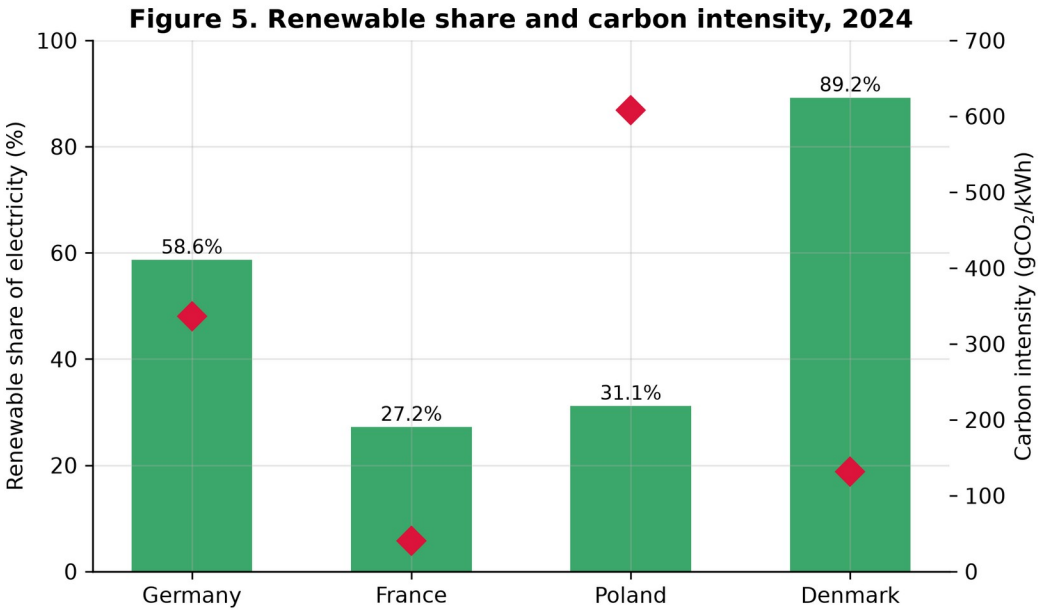


Figure 5. Renewable share and carbon intensity, 2024.

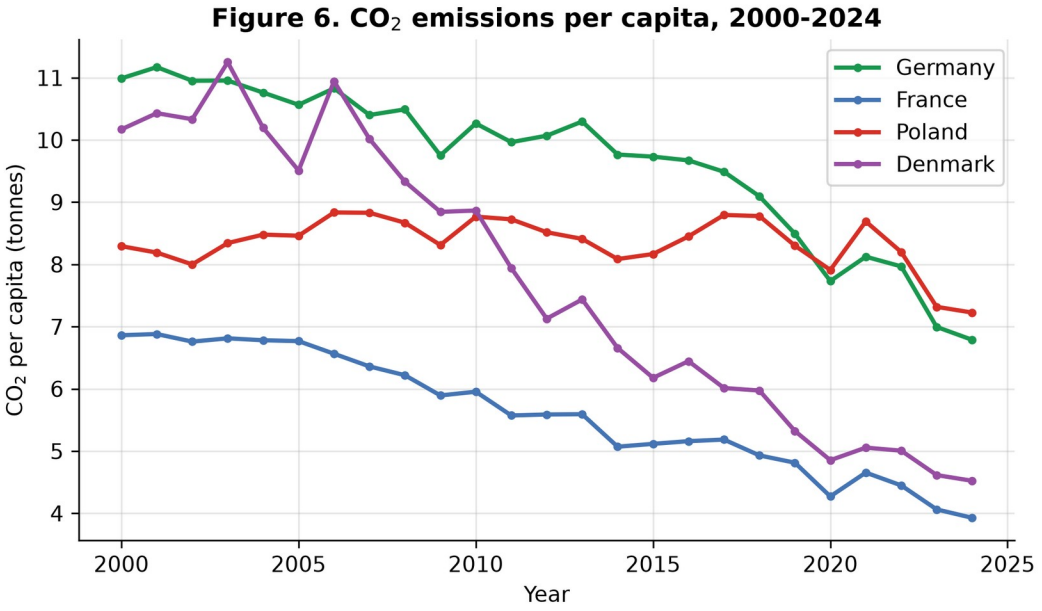


Figure 6. CO_2 emissions per capita, 2000–2024.

4. Discussion

Three points deserve emphasis. First, the German transition is real and large, but its *pace* is the policy problem: a trend that was sufficient

to reach 59% in 24 years is arithmetically insufficient for 80% by 2030. Second, the composition matters. Because wind and solar supplied 80.8% of renewable growth, future progress depends on the two most weather-variable sources, which raises grid, storage and flexibility questions that generation shares alone cannot answer. Third, the comparison warns against equating "renewable" with "low-carbon" too quickly: France's nuclear-based system is far cleaner per kWh today, while Denmark shows that a renewables-led route can go furthest on per-capita emissions. The 2022 coal rebound adds a resilience lesson: progress built over two decades survived a severe supply shock with only a one-year interruption — evidence that the transition is structural, not cyclical.

5. Limitations

This is a descriptive analysis of annual aggregates, and four limits apply. (1) The regression of carbon intensity on the renewable share is an *association*: demand changes, fuel prices, carbon pricing and plant closures move both variables, so the slope must not be read as the causal effect of one percentage point of renewables. (2) Annual data hide within-year variability — hourly matching of wind/solar output to demand is where integration costs appear. (3) Shares are computed on gross generation; cross-country differences in electricity trade (e.g. Denmark's interconnection) are not modelled. (4) The 2039 projection is a deliberately naive linear extrapolation used to gauge the distance to

the 2030 target, not a forecast. Extending the study with hourly data, a multivariate model, and cost data are natural next steps.

6. Conclusions

Between 2000 and 2024 Germany moved from 6.2% to 58.6% renewable electricity, ended nuclear generation, cut coal's share by more than half, and reduced the carbon intensity of its electricity by 41.3% and CO_2 per capita by 38.2%. Wind and solar drove the change; two major shocks disturbed but did not reverse it. The remaining gap — phasing out the last fifth of coal generation and accelerating from a +2.05 points-per-year trend to reach 80% by 2030 — is the central empirical fact any continuation of this work, or of the policy itself, must start from.

Data Availability

Analysis code, the extracted dataset, SQL queries and all figures: `github.com/rashedrana/germany-renewable-energy-analysis` (repository of this project). Source datasets: OWID energy data (github.com/owid/energy-data) and OWID CO_2 data (github.com/owid/co2-data), both CC BY.

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Declaration

This draft was prepared with AI assistance for data analysis and drafting, under the author's direction; all results were verified against the published datasets and the analysis code above, and the author takes full responsibility for the content. The disclosure will be adapted to the policy of the venue at submission.